# PCS External Communication Protocol

Applicable to IES9xx Series 1725 kW to 3.5 MW PCS

| Version | Date | Description | Revised By | Reviewed By |
|---|---|---|---|---|
| 1.0 | June 24, 2024 | First release. | 10034153 | |
| 1.1 | July 1, 2024 | Modified the precision of related parameters such as the current and power in the remote regulation, remote measurement, and GOOSE data to 1. | 10034153 | |
| 1.2 | July 19, 2024 | 1. Corrected parameter description of the orthogonal DC circuit breaker.<br>2. Added parameter D54 "PCS system status–value mode". | 10034153 | |
| 1.3 | August 9, 2024 | Changed parameter P03 for GOOSE remote measurement to "PCS system status–value mode". Adapted to primary ARMs V3.86 and later. | 10004590 | |
| 1.4 | August 23, 2024 | Changed the type of remote regulation data from Int to Float. | 10034153 | |
| 1.5 | August 30, 2024 | 1. Changed the minimum value of A16 to 0, the maximum value of A17 to 200, the default value of A19 to 0, and the default value of A20 to 5.<br>2. Added the remote measurement parameter D55 "Internal cavity temperature". | 10034153 | |
| 1.6 | October 8, 2024 | 1. Added the following remote measurement parameters: "AC charge phase A/B/C current" and "AC discharge phase A/B/C current".<br>2. Added the following remote signaling parameters: "Charge lockout status" and "Discharge lockout status". | 10034153 | |
| 1.7 | October 10, 2024 | Added the following faults or alarms: "Control mode inconsistent", "Capacitor harmonic overlimit", and "Fan pump and water pump switch error". | 10034153 | |
| 1.8 | November 25, 2024 | Added the following fault: "PCS association fault". | 10004590 | |
| 1.9 | December 6, 2024 | 1. Changed the upper limit of parameter A03 "Off-grid line voltage RMS" to 759.<br>2. Changed the upper limit of parameter A04 "PCS on-grid control mode" to 2.<br>3. Changed A12 "Rated grid frequency setting" to "Reserved".<br>4. Modified the name of parameters A15 and A18. | 10034153 | |
| 2.0 | December 11, 2024 | 1. Deleted the remote regulation parameter "Inertia response time constant (reserved)".<br>2. Added remote regulation parameters "Full-discharge scheduling function selection" and "Full-discharge scheduling time setting". | 10004590 | |
| 2.1 | December 25, 2024 | Added fault status J09 "Capacitor harmonic sampling disconnected". | 10034153 | |
| 2.2 | March 13, 2025 | 1. Changed the remote control parameter "Grid frequency adaptive control selection" to "Reserved". | 10034153 | |

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|---|---|---|---|---|
| | | 2. Added parameter G06 "Phase lock error" to fault status 2.<br>3. Added parameter L06 "Phase lock error" to alarm status 2. | | |

# Content

**Content** ..... **4**

**1 Overview** ..... **5**

**2 Control Parameters (CTRL)** ..... **5**

2.1 Remote Regulation Parameters (setGGIO1) ..... 5

2.2 Remote Control Parameters (ctlGAPC1) ..... 6

**3 Measurement Parameters (MEAS)** ..... **6**

3.1 Remote Signaling Parameters (staGGIO1) ..... 6

3.2 Remote Measurement Parameters (measGGIO1) ..... 7

3.3 BMS Parameters (measGGIO2) ..... 10

3.4 Fault Status 1 (alarmGGIO1) ..... 11

3.5 Fault Status 2 (alarmGGIO2) ..... 12

3.6 Fault Status 3 (alarmGGIO3) ..... 13

3.7 Fault Status 4 (alarmGGIO4) ..... 14

3.8 Fault Status 5 (alarmGGIO5) ..... 15

3.9 Alarm Status 1 (warningGGIO1) ..... 16

3.10 Alarm Status 2 (warningGGIO2) ..... 17

3.11 Alarm Status 3 (warningGGIO3) ..... 18

3.12 Alarm Status 4 (warningGGIO4) ..... 19

3.13 Alarm Status 5 (warningGGIO5) ..... 20

**4 GOOSE Data (PIGO)** ..... **21**

4.1 Remote Measurement Parameters (measGGIO1) ..... 21

# 1 Overview

This protocol applies to communication between the **IES9xx series** PCS from Wattox and monitoring software on the IEC 61850 host controller. This protocol enables real-time reading of PCS operational data, related statuses, and control command settings.

# 2 Control Parameters (CTRL)

## 2.1 Remote Regulation Parameters (setGGIO1)

| Param. | Name | Data Type | Description | Lower Limit | Upper Limit | Default | Unit |
|---|---|---|---|---|---|---|---|
| A01 | Active power setting | Float32 | - | –32768 | 32767 | 0 | 1 kW |
| A02 | Reactive power setting | Float32 | - | –32768 | 32767 | 0 | 1 kVar |
| A03 | Off-grid line voltage RMS | Float32 | | 0.0 | 759.0 | 690.0 | 0.1 V |
| A04 | PCS on-grid control mode | Float32 | 0: P-Q mode<br>1: Constant DC current mode<br>2: Constant DC voltage mode | 0 | 2 | 0 | 1 |
| A05 | Power change rate setting | Float32 | - | 10 | 30000 | 10000 | 1% Pn/s |
| A06 | Constant power factor setting | Float32 | - | –1.000 | 1.000 | 1.000 | 0.001 |
| A07 | DC bus voltage setting | Float32 | - | 0 | 1500.0 | 1300.0 | 0.1 V |
| A08 | Current setting in constant current mode | Float32 | Sets the bus current (actual value) for constant-current charge/discharge in P-Q mode. | –32768 | 32767 | 0 | 1 A |
| A09 | PCS control command | Float32 | 0: Off<br>1: On<br>2: Reset<br>3: Standby | 0 | 3 | 0 | 1 |
| A10 | PCS operation mode | Float32 | 0: On-grid<br>1: Off-grid | 0 | 1 | 0 | 1 |
| A11 | MMS and GOOSE fast control priority | Float32 | 0: None<br>1: GOOSE valid<br>2: MMS valid | 0 | 2 | 0 | 1 |
| A12 | Reserved | | | | | | |
| A13 | Reactive power mode setting | Float32 | 0: Constant reactive power<br>1: Constant PF<br>2: Q-U reactive | 0 | 2 | 0 | |
| A14 | Grid-forming function enable | Float32 | 0: Disable<br>1: Enable | 0 | 1 | 0 | 1 |
| A15 | Primary frequency control enable | Float32 | 0: Disable<br>1: Enable | 0 | 1 | 0 | 1 |
| A16 | Primary frequency control dead zone | Float32 | | 0 | 0.1 | 0.03 | 0.01 Hz |
| A17 | Primary frequency control coefficient | Float32 | | 0 | 65535 | 50 | 1 |
| A18 | Primary voltage control enable | Float32 | 0: Disable<br>1: Enable | 0 | 1 | 0 | 1 |
| A19 | Primary voltage control dead zone | Float32 | | 0 | 1.00 | 0.0 | 0.01 |
| A20 | Primary voltage control coefficient | Float32 | | 0 | 100.0 | 1.0 | 0.1 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| A21 | Virtual inertia coefficient | Float32 | | 0 | 65535 | 2000 | 1 |
| A22 | Virtual damping coefficient | Float32 | | 0 | 65535 | 1500 | 1 |
| A23 | Virtual excitation coefficient | Float32 | | 0 | 65535 | 1200 | 1 |
| A24 | Reserved | Float32 | | 0 | 65535 | 0 | 1 |
| A25 | Full-discharge scheduling function selection | Float32 | 0: Disable<br>1: Method 1 | 0 | 1 | 0 | 1 |
| A26 | Full-discharge scheduling time setting | Float32 | | 0 | 65535 | 0 | 1 |

*: A11 priority setting only applies to PCS control commands (on/off/standby), active power setting, and reactive power setting.

### 2.2 Remote Control Parameters (ctlGAPC1)

| Param. | Name | Data Type | Description | Lower Limit | Upper Limit | Default | Unit |
|---|---|---|---|---|---|---|---|
| B01 | Reserved | BOOL | - | 0 | 1 | 0 | 1 |
| B02 | PCS on/off | BOOL | False: Off<br>True: On | 0 | 1 | 0 | 1 |
| B03 | Start command | BOOL | False: Invalid<br>True: Valid | 0 | 1 | 0 | 1 |
| B04 | Stop command | BOOL | False: Invalid<br>True: Valid | 0 | 1 | 0 | 1 |
| B05 | Fault reset | BOOL | False: Invalid<br>True: Valid | 0 | 1 | 0 | 1 |
| B06 | Standby command | BOOL | False: Invalid<br>True: Valid | 0 | 1 | 0 | 1 |

## 3 Measurement Parameters (MEAS)

### 3.1 Remote Signaling Parameters (staGGIO1)

| Param. | Name | Data Type | Description | Lower Limit | Upper Limit | Default | Unit |
|---|---|---|---|---|---|---|---|
| C01 | AC circuit breaker status | BOOL | False: Closed<br>True: Open | 0 | 1 | 0 | 1 |
| C02 | DC disconnector status (module 1) | BOOL | False: Closed<br>True: Open | 0 | 1 | 0 | 1 |
| C03 | DC disconnector status (module 2) | BOOL | False: Closed<br>True: Open | 0 | 1 | 0 | 1 |
| C04 | AC pre-charge contactor status | BOOL | False: Closed<br>True: Open | 0 | 1 | 0 | 1 |
| C05 | DC pre-charge contactor status (module 1) | BOOL | False: Closed<br>True: Open | 0 | 1 | 0 | 1 |
| C06 | DC pre-charge contactor status (module 2) | BOOL | False: Closed<br>True: Open | 0 | 1 | 0 | 1 |
| C07 | Emergency stop button status | BOOL | False: Normal<br>True: Emergency stop | 0 | 1 | 0 | 1 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| C08 | PCS on/off status | BOOL | False: Off<br>True: On | 0 | 1 | 0 | 1 |
| C09 | Command source | BOOL | False: Local<br>True: Remote | 0 | 1 | 0 | 1 |
| C10 | On-grid charging operation status | BOOL | False: Non-charging operation<br>True: Charging operation | 0 | 1 | 0 | 1 |
| C11 | On-grid discharging operation status | BOOL | False: Non-discharging operation<br>True: Discharging operation | 0 | 1 | 0 | 1 |
| C12 | On-grid total operation status | BOOL | False: Non-operating<br>True: Operating | 0 | 1 | 0 | 1 |
| C13 | Standby status | BOOL | False: Non-standby True: Standby | 0 | 1 | 0 | 1 |
| C14 | Stop status | BOOL | False: Non-stopped True: Stopped | 0 | 1 | 0 | 1 |
| C15 | Fault status | BOOL | False: No fault True: Fault occurred | 0 | 1 | 0 | 1 |
| C16 | Alarm status | BOOL | False: No alarm True: Alarm occurred | 0 | 1 | 0 | 1 |
| C17 | Off-grid operation status | BOOL | False: Non-off-grid state True: Off-grid operation | 0 | 1 | 0 | 1 |
| C18 | Lockout status | BOOL | False: None<br>True: Lockout active | 0 | 1 | 0 | 1 |
| C19 | Charge lockout status | BOOL | False: None<br>True: Charge lockout | 0 | 1 | 0 | 1 |
| C20 | Discharge lockout status | BOOL | False: None<br>True: Discharge lockout | 0 | 1 | 0 | 1 |

## 3.2 Remote Measurement Parameters (measGGIO1)

| Param. | Name | Data Type | Description | Lower Limit | Upper Limit | Default | Unit |
|---|---|---|---|---|---|---|---|
| D01 | Grid AB line voltage RMS | Float32 | - | 0 | 6553.5 | 0 | 0.1 V |
| D02 | Grid BC line voltage RMS | Float32 | - | 0 | 6553.5 | 0 | 0.1 V |
| D03 | Grid CA line voltage RMS | Float32 | - | 0 | 6553.5 | 0 | 0.1 V |
| D04 | Grid A-phase current RMS | Float32 | - | 0 | 65535 | 0 | 1 A |
| D05 | Grid B-phase current RMS | Float32 | - | 0 | 65535 | 0 | 1 A |
| D06 | Grid C-phase current RMS | Float32 | - | 0 | 65535 | 0 | 1 A |
| D07 | Total active output power | Float32 | - | –32768 | 32767 | 0 | 1 kW |
| D08 | Total reactive output power | Float32 | - | –32768 | 32767 | 0 | 1 kVar |
| D09 | Output power factor | Float32 | - | –1.000 | 1.000 | 0 | 0.001 |
| D10 | Battery power | Float32 | - | –32768 | 32767 | 0 | 1 kW |
| D11 | Grid apparent power | Float32 | - | –32768 | 32767 | 0 | 1 kVA |
| D12 | Current maximum allowable discharging power | Float32 | - | 0 | 65535 | 0 | 1 kW |
| D13 | Current maximum allowable charging power | Float32 | - | 0 | 65535 | 0 | 1 kW |
| D14 | Maximum charging power of battery pack | Float32 | | 0 | 65535 | 0 | 1 kW |
| D15 | Maximum discharging power of battery pack | Float32 | - | 0 | 65535 | 0 | 1 kW |
| D16 | DC side charging current | Float32 | - | 0 | 65535 | 0 | 1 A |
| D17 | DC side discharging current | Float32 | - | 0 | 65535 | 0 | 1 A |
| D18 | DC side charging power | Float32 | | 0 | 65535 | 0 | 1 kW |
| D19 | DC side discharging power | Float32 | - | 0 | 65535 | 0 | 1 kW |
| D20 | AC side total charge active power | Float32 | - | 0 | 65535 | 0 | 1 kW |
| D21 | AC side total discharge active power | Float32 | - | 0 | 65535 | 0 | 1 kW |
| D22 | AC side total charge reactive power | Float32 | - | 0 | 65535 | 0 | 1 kW |
| D23 | AC side total discharge reactive power | Float32 | - | 0 | 65535 | 0 | 1 kW |
| D24 | Daily discharge amount | Float32 | Reset at midnight | 0 | 65535 | 0 | 1 kWh |
| D25 | Daily charge amount | Float32 | Reset at midnight | 0 | 65535 | 0 | 1 kWh |
| D27 | All discharge energy | Float32 | | 0 | - | 0 | 1 kWh |
| D29 | All charge energy | Float32 | - | 0 | - | 0 | 1 kWh |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| D30 | PCS system status–bit mode | Int32 | Bit 0: Stop<br>Bit 1: Standby<br>Bit 2: Grid-connected charging operation<br>Bit 3: Grid-connected discharging operation<br>Bit 4: Grid-connected zero power operation<br>Bit 5: Off-grid operation<br>Bit 6: Fault status<br>Bit 7: Alarm status | 0 | 65535 | 0 | 1 |
| D31 | A-phase NTC temperature (module 1) | Float32 | - | –3276.8 | 3276.7 | 0 | 0.1°C |
| D32 | B-phase NTC temperature (module 1) | Float32 | - | –3276.8 | 3276.7 | 0 | 0.1°C |
| D33 | C-phase NTC temperature (module 1) | Float32 | - | –3276.8 | 3276.7 | 0 | 0.1°C |
| D34 | Reactor temperature (module 1) | Float32 | | –3276.8 | 3276.7 | 0 | 0.1°C |
| D35 | A-phase NTC temperature (module 2) | Float32 | - | –3276.8 | 3276.7 | 0 | 0.1°C |
| D36 | B-phase NTC temperature (module 2) | Float32 | - | –3276.8 | 3276.7 | 0 | 0.1°C |
| D37 | C-phase NTC temperature (module 2) | Float32 | - | –3276.8 | 3276.7 | 0 | 0.1°C |
| D38 | Reactor temperature (module 2) | Float32 | | –3276.8 | 3276.7 | 0 | 0.1°C |
| D39 | Ambient temperature | Float32 | | –3276.8 | 3276.7 | 0 | 0.1°C |
| D40 | System insulation impedance | Float32 | | 0 | 65535 | 0 | 1 kΩ |
| D41 | Battery voltage | Float32 | | –3276.8 | 3276.7 | 0 | 0.1 V |
| D42 | DC current | Float32 | | –32768 | 32767 | 0 | 1 A |
| D43 | Upper half bus voltage | Float32 | | –3276.8 | 3276.7 | 0 | 0.1 V |
| D44 | Lower half bus voltage | Float32 | | –3276.8 | 3276.7 | 0 | 0.1 V |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| D45 | Grid frequency | Float32 | | 0 | 655.35 | 0 | 0.01 Hz |
| D46 | Rated power | Float32 | | 0 | 65535 | 1725 | 1 |
| D47 | Reactive power upper limit | Float32 | | 0 | 32767 | 0 | 1 |
| D48 | Reactive power lower limit | Float32 | | –32768 | 0 | 0 | 1 |
| D49 | Derating flag | Int32 | Bit 0: Derating due to ambient temperature<br>Bit 1: Derating due to coolant inlet temperature<br>Bit 2: Derating due to fan<br>Bit 3: Derating due to reactor temperature<br>Bit 4: Master and slave derating together | 0 | 65535 | 0 | 1 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| D50 | DI status (module 1) | Int32 | Bit 0: Door status (0: Closed; 1: Open)<br>Bit 1: Insulation detection contactor status (0: Closed; 1: Open)<br>Bit 2: Water level detection status (0: Normal; 1: Fault)<br>Bit 3: DC surge protection device status (0: Normal; 1: Fault)<br>Bit 4: AC surge protection device status (0: Normal; 1: Fault)<br>Bit 5: Water pump power contactor status (0: Closed; 1: Open)<br>Bit 6: Fan power contactor status (0: Closed; 1: Open)<br>Bit 7: Heater contactor status (0: Closed; 1: Open)<br>Bit 8: Reactor temperature control switch status (0: Normal temperature; 1: Over-temperature)<br>Bit 9: Reserved for DI9<br>Bit 10: Reserved for DI11<br>Bit 11: Reserved for DI12<br>Bit 12: Reserved for DI13<br>Bit 13: Reserved for DI14<br>Bit 14: Reserved for DI15<br>Bit 15: Reserved for DI16 | 0 | 65535 | 0 | 1 |
| D51 | DI status (module 2) | Int32 | Same as above | 0 | 65535 | 0 | 1 |
| D52 | Grid-forming PCS internal frequency (reserved) | Float32 | - | 40.00 | 55.00 | 50.00 | 0.01 Hz |
| D53 | Grid-forming PCS virtual potential (reserved) | Float32 | - | 0.0 | 750.0 | 565.0 | 0.1 V |
| D54 | PCS system status–value mode | Int32 | 0: Stopped<br>1: Standby<br>2: On-grid charging operation<br>3: On-grid discharging operation<br>4: On-grid zero power operation<br>5: Off-grid operation<br>6: Fault status | 0 | 65535 | 0 | 1 |
| D55 | Internal cavity temperature | Float32 | | –3276.8 | 3276.7 | 0 | 0.1°C |
| D56 | Grid charge A-phase current RMS | Float32 | - | 0 | 65535 | 0 | 1 A |
| D57 | Grid charge B-phase current RMS | Float32 | - | 0 | 65535 | 0 | 1 A |
| D58 | Grid charge C-phase current RMS | Float32 | - | 0 | 65535 | 0 | 1 A |
| D59 | Grid discharge A-phase current RMS | Float32 | - | 0 | 65535 | 0 | 1 A |
| D60 | Grid discharge B-phase current RMS | Float32 | - | 0 | 65535 | 0 | 1 A |
| D61 | Grid discharge C-phase current RMS | Float32 | - | 0 | 65535 | 0 | 1 A |

## 3.3 BMS Parameters (measGGIO2)

| Param. | Name | Data Type | Description | Lower Limit | Upper Limit | Default | Unit |
|---|---|---|---|---|---|---|---|
| E01 | Battery pack SOC | Float32 | - | 0 | 100.0 | - | 0.1% |
| E02 | Maximum charging power of battery pack | Float32 | - | 0 | 6553.5 | - | 0.1 kW |
| E03 | Maximum discharging power of battery pack | Float32 | - | 0 | 6553.5 | - | 0.1 kW |
| E04 | BMS operation status | Int32 | 00: Initial<br>01: Normal<br>02: Charge prohibited<br>03: Discharge prohibited<br>04: Alarm<br>05: Standby<br>06: Fault<br>07: Reserved | 0 | 65535 | - | 1 |
| E05 | BMS status word | Int32 | See Table 1. | 0 | 65535 | - | 1 |
| E06 | Total voltage of battery pack | Float32 | - | 0 | 6553.5 | - | 0.1 V |
| E07 | Total current of battery pack | Float32 | The value is negative during charging and positive during discharging. | –3276.8 | 3276.7 | - | 0.1 A |
| E08 | Maximum charging current of battery pack | Float32 | - | 0 | 6553.5 | - | 0.1 A |
| E09 | Maximum discharging current of battery pack | Float32 | - | 0 | 6553.5 | - | 0.1 A |
| E10 | Highest cell voltage | Float32 | - | 0 | 65.535 | - | 0.001 V |
| E11 | Lowest cell voltage | Float32 | - | 0 | 65.535 | - | 0.001 V |
| E12 | Battery pack SOH | Float32 | - | 0 | 100.0 | - | 0.1% |
| E13 | Reserved | | | | | | |
| E14 | Reserved | | | | | | |

**Table 1 BMS status word**

| Name | Bit | Parameter | Description |
|---|---|---|---|
| **BMS status word** | 0 | BMS no communication | 0: Invalid; 1: Occurred |
| | 1 | BMS fault | 0: Invalid; 1: Occurred |
| | 2 | SOC above the upper limit | 0: Invalid; 1: Occurred |
| | 3 | SOC below the lower limit | 0: Invalid; 1: Occurred |
| | 4 | BMS charge prohibited | 0: Invalid; 1: Occurred |
| | 5 | BMS discharge prohibited | 0: Invalid; 1: Occurred |
| | 6 | BMS initialization | 0: Invalid; 1: Occurred |
| | 7 | BMS alarm | 0: Invalid; 1: Occurred |
| | 8 | BMS charge and discharge prohibited | 0: Invalid; 1: Occurred |
| | 9 | Reserved | - |
| | 10 | Reserved | - |
| | 11 | Reserved | - |
| | 12 | Reserved | - |
| | 13 | Reserved | - |
| | 14 | Reserved | - |
| | 15 | Reserved | - |

[1] When the BMS status word value is 0 (that is, all bits are 0), it indicates that the **BMS system is normal**.

### 3.4 Fault Status 1 (alarmGGIO1)

| Param. | Fault Name | Data Type | Description |
|---|---|---|---|
| F01 | Storage battery polarity connected reversely | BOOL | 0: No fault; 1: Fault occurred |
| F02 | Storage battery voltage too high | BOOL | 0: No fault; 1: Fault occurred |
| F03 | Storage battery voltage too low | BOOL | 0: No fault; 1: Fault occurred |
| F04 | Battery voltage too high | BOOL | 0: No fault; 1: Fault occurred |
| F05 | Battery voltage too low | BOOL | 0: No fault; 1: Fault occurred |
| F06 | BMS system fault | BOOL | 0: No fault; 1: Fault occurred |
| F07 | BMS pre-charge fault | BOOL | 0: No fault; 1: Fault occurred |
| F08 | BMS main contactor fault | BOOL | 0: No fault; 1: Fault occurred |
| F09 | BMS insulation fault | BOOL | 0: No fault; 1: Fault occurred |
| F10 | DC side half-bus hardware overvoltage | BOOL | 0: No fault; 1: Fault occurred |
| F11 | DC side full-bus software overvoltage | BOOL | 0: No fault; 1: Fault occurred |
| F12 | DC side full-bus software undervoltage | BOOL | 0: No fault; 1: Fault occurred |

| | | | |
|---|---|---|---|
| F13 | DC side half-bus voltage variation too large | BOOL | 0: No fault; 1: Fault occurred |
| F14 | AC side hardware overcurrent | BOOL | 0: No fault; 1: Fault occurred |
| F15 | AC side software overcurrent | BOOL | 0: No fault; 1: Fault occurred |
| F16 | DC side hardware overcurrent | BOOL | 0: No fault; 1: Fault occurred |
| F17 | DC side software overcurrent | BOOL | 0: No fault; 1: Fault occurred |
| F18 | IGBT Desat fault | BOOL | 0: No fault; 1: Fault occurred |
| F19 | IGBT block fault | BOOL | 0: No fault; 1: Fault occurred |
| F20 | IGBT over-temperature | BOOL | 0: No fault; 1: Fault occurred |
| F21 | AC side A-phase Hall disconnected | BOOL | 0: No fault; 1: Fault occurred |
| F22 | AC side B-phase Hall disconnected | BOOL | 0: No fault; 1: Fault occurred |
| F23 | AC side C-phase Hall disconnected | BOOL | 0: No fault; 1: Fault occurred |
| F24 | DC side current Hall disconnected | BOOL | 0: No fault; 1: Fault occurred |
| F25 | Detection board model mismatch | BOOL | 0: No fault; 1: Fault occurred |
| F26 | DC side pre-charge contactor fault | BOOL | 0: No fault; 1: Fault occurred |
| F27 | DC side switch fault | BOOL | 0: No fault; 1: Fault occurred |
| F28 | AC side pre-charge contactor fault | BOOL | 0: No fault; 1: Fault occurred |
| F29 | AC side switch fault | BOOL | 0: No fault; 1: Fault occurred |
| F30 | CBC overcurrent | BOOL | 0: No fault; 1: Fault occurred |
| F31 | Device overload | BOOL | 0: No fault; 1: Fault occurred |
| F32 | Grid AC overvoltage | BOOL | 0: No fault; 1: Fault occurred |

### 3.5 Fault Status 2 (alarmGGIO2)

| Param. | Fault Name | Data Type | Description |
|---|---|---|---|
| G01 | Grid AC undervoltage | BOOL | 0: No fault; 1: Fault occurred |
| G02 | Grid AC overfrequency | BOOL | 0: No fault; 1: Fault occurred |
| G03 | Grid AC underfrequency | BOOL | 0: No fault; 1: Fault occurred |
| G04 | Output phase loss | BOOL | 0: No fault; 1: Fault occurred |
| G05 | Grid voltage sequence incorrect | BOOL | 0: No fault; 1: Fault occurred |
| G06 | Phase lock error | BOOL | 0: No fault; 1: Fault occurred |
| G07 | Grid voltage imbalance | BOOL | 0: No fault; 1: Fault occurred |
| G08 | AC overvoltage in off-grid mode | BOOL | 0: No fault; 1: Fault occurred |
| G09 | AC undervoltage in off-grid mode | BOOL | 0: No fault; 1: Fault occurred |
| G10 | Islanding fault | BOOL | 0: No fault; 1: Fault occurred |
| G11 | Current control deviation too large | BOOL | 0: No fault; 1: Fault occurred |

| | | | |
|---|---|---|---|
| G12 | Bus soft-start failure | BOOL | 0: No fault; 1: Fault occurred |
| G13 | DC component in the current beyond limit on the AC side | BOOL | 0: No fault; 1: Fault occurred |
| G14 | AC switch closing voltage mismatch | BOOL | 0: No fault; 1: Fault occurred |
| G15 | AD zero drift too large | BOOL | 0: No fault; 1: Fault occurred |
| G16 | Low insulation resistance | BOOL | 0: No fault; 1: Fault occurred |
| G17 | Insulation detection failed | BOOL | 0: No fault; 1: Fault occurred |
| G18 | Internal cavity over-temperature | BOOL | 0: No fault; 1: Fault occurred |
| G19 | Internal cavity over-temperature alarm | BOOL | 0: No fault; 1: Fault occurred |
| G20 | Internal cavity temperature low | BOOL | 0: No fault; 1: Fault occurred |
| G21 | Filter capacitor over-temperature | BOOL | 0: No fault; 1: Fault occurred |
| G22 | Filter capacitor over-temperature alarm | BOOL | 0: No fault; 1: Fault occurred |
| G23 | Filter capacitor temperature low | BOOL | 0: No fault; 1: Fault occurred |
| G24 | Auxiliary transformer over-temperature | BOOL | 0: No fault; 1: Fault occurred |
| G25 | Auxiliary transformer over-temperature alarm | BOOL | 0: No fault; 1: Fault occurred |
| G26 | Auxiliary transformer temperature low | BOOL | 0: No fault; 1: Fault occurred |
| G27 | High ambient temperature | BOOL | 0: No fault; 1: Fault occurred |
| G28 | High ambient temperature alarm | BOOL | 0: No fault; 1: Fault occurred |
| G29 | Low ambient temperature | BOOL | 0: No fault; 1: Fault occurred |
| G30 | IGBT module A-phase over-temperature | BOOL | 0: No fault; 1: Fault occurred |
| G31 | IGBT module B-phase over-temperature | BOOL | 0: No fault; 1: Fault occurred |
| G32 | IGBT module C-phase over-temperature | BOOL | 0: No fault; 1: Fault occurred |

## 3.6 Fault Status 3 (alarmGGIO3)

| Param. | Fault Name | Data Type | Description |
|---|---|---|---|
| H01 | IGBT temperature sampling error | BOOL | 0: No fault; 1: Fault occurred |
| H02 | IGBT over-temperature alarm | BOOL | 0: No fault; 1: Fault occurred |
| H03 | Reactor over-temperature | BOOL | 0: No fault; 1: Fault occurred |
| H04 | Reactor over-temperature alarm | BOOL | 0: No fault; 1: Fault occurred |
| H05 | Reactor temperature low | BOOL | 0: No fault; 1: Fault occurred |
| H06 | Surge protection device fault | BOOL | 0: No fault; 1: Fault occurred |
| H07 | Parallel communication fault | BOOL | 0: No fault; 1: Fault occurred |
| H08 | Abnormal stop of the master | BOOL | 0: No fault; 1: Fault occurred |
| H09 | Carrier synchronization failed | BOOL | 0: No fault; 1: Fault occurred |
| H10 | Internal cavity fan fault | BOOL | 0: No fault; 1: Fault occurred |
| H11 | Circulating fan fault | BOOL | 0: No fault; 1: Fault occurred |
| H12 | Reactor fan fault | BOOL | 0: No fault; 1: Fault occurred |
| H13 | IGBT fan fault | BOOL | 0: No fault; 1: Fault occurred |
| H14 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| H15 | CAN communication fault between control board and communication board | BOOL | 0: No fault; 1: Fault occurred |
| H16 | HMI communication fault | BOOL | 0: No fault; 1: Fault occurred |
| H17 | BMS communication fault | BOOL | 0: No fault; 1: Fault occurred |
| H18 | EMS connection timeout 1 (external bus) | BOOL | 0: No fault; 1: Fault occurred |
| H19 | EMS connection timeout 1 (internal communication) | BOOL | 0: No fault; 1: Fault occurred |
| H20 | EMS connection timeout 2 (external bus) | BOOL | 0: No fault; 1: Fault occurred |
| H21 | EMS connection timeout 2 (internal communication) | BOOL | 0: No fault; 1: Fault occurred |
| H22 | EMS communication fault (dual network) | BOOL | 0: No fault; 1: Fault occurred |
| H23 | EEPROM fault | BOOL | 0: No fault; 1: Fault occurred |
| H24 | EEPROM alarm | BOOL | 0: No fault; 1: Fault occurred |
| H25 | Black box error | BOOL | 0: No fault; 1: Fault occurred |
| H26 | FPGA data exchange error | BOOL | 0: No fault; 1: Fault occurred |
| H27 | RTC error | BOOL | 0: No fault; 1: Fault occurred |
| H28 | 220 V control power disconnected | BOOL | 0: No fault; 1: Fault occurred |
| H29 | Leakage current Hall self-test failed | BOOL | 0: No fault; 1: Fault occurred |
| H30 | Residual abrupt leakage current overlimit | BOOL | 0: No fault; 1: Fault occurred |
| H31 | Continuous residual leakage current overlimit | BOOL | 0: No fault; 1: Fault occurred |
| H32 | Local emergency stop fault | BOOL | 0: No fault; 1: Fault occurred |

## 3.7 Fault Status 4 (alarmGGIO4)

| Param. | Fault Name | Data Type | Description |
|---|---|---|---|
| I01 | Remote emergency stop fault | BOOL | 0: No fault; 1: Fault occurred |
| I02 | BMS emergency stop fault | BOOL | 0: No fault; 1: Fault occurred |
| I03 | Phase-to-PE fault | BOOL | 0: No fault; 1: Fault occurred |
| I04 | Capacitor voltage resonance fault | BOOL | 0: No fault; 1: Fault occurred |
| I05 | High humidity | BOOL | 0: No fault; 1: Fault occurred |
| I06 | Humidity detection disconnected | BOOL | 0: No fault; 1: Fault occurred |
| I07 | Temperature too high | BOOL | 0: No fault; 1: Fault occurred |
| I08 | Temperature detection disconnected | BOOL | 0: No fault; 1: Fault occurred |
| I09 | Fuse fault | BOOL | 0: No fault; 1: Fault occurred |
| I10 | Cabinet door fault | BOOL | 0: No fault; 1: Fault occurred |
| I11 | Water pump locked-rotor | BOOL | 0: No fault; 1: Fault occurred |
| I12 | Water pump speed regulation signal absent | BOOL | 0: No fault; 1: Fault occurred |
| I13 | Water pump idling | BOOL | 0: No fault; 1: Fault occurred |
| I14 | Water cooling system hydraulic fault | BOOL | 0: No fault; 1: Fault occurred |
| I15 | Fan power sequence incorrect | BOOL | 0: No fault; 1: Fault occurred |
| I16 | Fan power disconnected | BOOL | 0: No fault; 1: Fault occurred |
| I17 | Three-phase fan fault | BOOL | 0: No fault; 1: Fault occurred |
| I18 | Heater contactor fault | BOOL | 0: No fault; 1: Fault occurred |
| I19 | Water pump power disconnected | BOOL | 0: No fault; 1: Fault occurred |
| I20 | Software version inconsistent between master and slave | BOOL | 0: No fault; 1: Fault occurred |
| I21 | Dynamic environment fault | BOOL | 0: No fault; 1: Fault occurred |
| I22 | DC side midpoint sampling disconnected | BOOL | 0: No fault; 1: Fault occurred |
| I23 | DC side voltage sampling disconnected | BOOL | 0: No fault; 1: Fault occurred |

| | | | |
|---|---|---|---|
| I24 | AC and DC voltage mismatch | BOOL | 0: No fault; 1: Fault occurred |
| I25 | Grid voltage mismatch | BOOL | 0: No fault; 1: Fault occurred |
| I26 | Heat exchanger fan fault | BOOL | 0: No fault; 1: Fault occurred |
| I27 | DC circuit breaker fan fault | BOOL | 0: No fault; 1: Fault occurred |
| I28 | AC circuit breaker fan fault | BOOL | 0: No fault; 1: Fault occurred |
| I29 | Master and slave identification fault | BOOL | 0: No fault; 1: Fault occurred |
| I30 | Power inconsistent between master and slave | BOOL | 0: No fault; 1: Fault occurred |
| I31 | IGBT temperature low | BOOL | 0: No fault; 1: Fault occurred |
| I32 | Zero sequence loop current too large | BOOL | 0: No fault; 1: Fault occurred |

### 3.8 Fault Status 5 (alarmGGIO5)

| Param. | Fault Name | Data Type | Description |
|---|---|---|---|
| J01 | FPGA side fault | BOOL | 0: No fault; 1: Fault occurred |
| J02 | Fault of parallel devices in cabinet | BOOL | 0: No fault; 1: Fault occurred |
| J03 | Fiber communication interrupted | BOOL | 0: No fault; 1: Fault occurred |
| J04 | Fiber communication CRC fault | BOOL | 0: No fault; 1: Fault occurred |
| J05 | Control mode inconsistent | BOOL | 0: No fault; 1: Fault occurred |
| J06 | Capacitor harmonic overlimit | BOOL | 0: No fault; 1: Fault occurred |
| J07 | Fan pump and water pump switch error | BOOL | 0: No fault; 1: Fault occurred |
| J08 | PCS interaction fault | BOOL | 0: No fault; 1: Fault occurred |
| J09 | Capacitor harmonic sampling disconnected | BOOL | 0: No fault; 1: Fault occurred |
| J10 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J11 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J12 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J13 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J14 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J15 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J16 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J17 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J18 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J19 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J20 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J21 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J22 | Reserved | BOOL | 0: No fault; 1: Fault occurred |

| | | | |
|---|---|---|---|
| J23 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J24 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J25 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J26 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J27 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J28 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J29 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J30 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J31 | Reserved | BOOL | 0: No fault; 1: Fault occurred |
| J32 | Reserved | BOOL | 0: No fault; 1: Fault occurred |

* Note: Fault statuses are continuously updated; for faults not listed, refer to HMI display or consult PCS manufacturer.

### 3.9 Alarm Status 1 (warningGGIO1)

| Param. | Fault Name | Data Type | Description |
|---|---|---|---|
| K01 | Storage battery polarity connected reversely | BOOL | 0: No alarm; 1: Alarm occurred |
| K02 | Storage battery voltage too high | BOOL | 0: No alarm; 1: Alarm occurred |
| K03 | Storage battery voltage too low | BOOL | 0: No alarm; 1: Alarm occurred |
| K04 | Battery voltage too high | BOOL | 0: No alarm; 1: Alarm occurred |
| K05 | Battery voltage too low | BOOL | 0: No alarm; 1: Alarm occurred |
| K06 | BMS system fault | BOOL | 0: No alarm; 1: Alarm occurred |
| K07 | BMS pre-charge fault | BOOL | 0: No alarm; 1: Alarm occurred |
| K08 | BMS main contactor fault | BOOL | 0: No alarm; 1: Alarm occurred |
| K09 | BMS insulation fault | BOOL | 0: No alarm; 1: Alarm occurred |
| K10 | DC side half-bus hardware overvoltage | BOOL | 0: No alarm; 1: Alarm occurred |
| K11 | DC side full-bus software overvoltage | BOOL | 0: No alarm; 1: Alarm occurred |
| K12 | DC side full-bus software undervoltage | BOOL | 0: No alarm; 1: Alarm occurred |
| K13 | DC side half-bus voltage variation too large | BOOL | 0: No alarm; 1: Alarm occurred |
| K14 | AC side hardware overcurrent | BOOL | 0: No alarm; 1: Alarm occurred |
| K15 | AC side software overcurrent | BOOL | 0: No alarm; 1: Alarm occurred |
| K16 | DC side hardware overcurrent | BOOL | 0: No alarm; 1: Alarm occurred |
| K17 | DC side software overcurrent | BOOL | 0: No alarm; 1: Alarm occurred |
| K18 | IGBT Desat fault | BOOL | 0: No alarm; 1: Alarm occurred |
| K19 | IGBT block fault | BOOL | 0: No alarm; 1: Alarm occurred |

| | | | |
|---|---|---|---|
| K20 | IGBT over-temperature | BOOL | 0: No alarm; 1: Alarm occurred |
| K21 | AC side A-phase Hall disconnected | BOOL | 0: No alarm; 1: Alarm occurred |
| K22 | AC side B-phase Hall disconnected | BOOL | 0: No alarm; 1: Alarm occurred |
| K23 | AC side C-phase Hall disconnected | BOOL | 0: No alarm; 1: Alarm occurred |
| K24 | DC side current Hall disconnected | BOOL | 0: No alarm; 1: Alarm occurred |
| K25 | Detection board model mismatch | BOOL | 0: No alarm; 1: Alarm occurred |
| K26 | DC side pre-charge contactor fault | BOOL | 0: No alarm; 1: Alarm occurred |
| K27 | DC side switch fault | BOOL | 0: No alarm; 1: Alarm occurred |
| K28 | AC side pre-charge contactor fault | BOOL | 0: No alarm; 1: Alarm occurred |
| K29 | AC side switch fault | BOOL | 0: No alarm; 1: Alarm occurred |
| K30 | CBC overcurrent | BOOL | 0: No alarm; 1: Alarm occurred |
| K31 | Device overload | BOOL | 0: No alarm; 1: Alarm occurred |
| K32 | Grid AC overvoltage | BOOL | 0: No alarm; 1: Alarm occurred |

### 3.10 Alarm Status 2 (warningGGIO2)

| Param. | Fault Name | Data Type | Description |
|---|---|---|---|
| L01 | Grid AC undervoltage | BOOL | 0: No alarm; 1: Alarm occurred |
| L02 | Grid AC overfrequency | BOOL | 0: No alarm; 1: Alarm occurred |
| L03 | Grid AC underfrequency | BOOL | 0: No alarm; 1: Alarm occurred |
| L04 | Output phase loss | BOOL | 0: No alarm; 1: Alarm occurred |
| L05 | Grid voltage sequence incorrect | BOOL | 0: No alarm; 1: Alarm occurred |
| L06 | Phase lock error | BOOL | 0: No alarm; 1: Alarm occurred |
| L07 | Grid voltage imbalance | BOOL | 0: No alarm; 1: Alarm occurred |
| L08 | AC overvoltage in off-grid mode | BOOL | 0: No alarm; 1: Alarm occurred |
| L09 | AC undervoltage in off-grid mode | BOOL | 0: No alarm; 1: Alarm occurred |
| L10 | Islanding fault | BOOL | 0: No alarm; 1: Alarm occurred |
| L11 | Current control deviation too large | BOOL | 0: No alarm; 1: Alarm occurred |
| L12 | Bus soft-start failure | BOOL | 0: No alarm; 1: Alarm occurred |
| L13 | DC component in the current beyond limit on the AC side | BOOL | 0: No alarm; 1: Alarm occurred |
| L14 | AC switch closing voltage mismatch | BOOL | 0: No alarm; 1: Alarm occurred |
| L15 | AD zero drift too large | BOOL | 0: No alarm; 1: Alarm occurred |
| L16 | Low insulation resistance | BOOL | 0: No alarm; 1: Alarm occurred |
| L17 | Insulation detection failed | BOOL | 0: No alarm; 1: Alarm occurred |
| L18 | Internal cavity over-temperature | BOOL | 0: No alarm; 1: Alarm occurred |

| | | | |
|---|---|---|---|
| L19 | Internal cavity over-temperature alarm | BOOL | 0: No alarm; 1: Alarm occurred |
| L20 | Internal cavity temperature low | BOOL | 0: No alarm; 1: Alarm occurred |
| L21 | Filter capacitor over-temperature | BOOL | 0: No alarm; 1: Alarm occurred |
| L22 | Filter capacitor over-temperature alarm | BOOL | 0: No alarm; 1: Alarm occurred |
| L23 | Filter capacitor temperature low | BOOL | 0: No alarm; 1: Alarm occurred |
| L24 | Auxiliary transformer over-temperature | BOOL | 0: No alarm; 1: Alarm occurred |
| L25 | Auxiliary transformer over-temperature alarm | BOOL | 0: No alarm; 1: Alarm occurred |
| L26 | Auxiliary transformer temperature low | BOOL | 0: No alarm; 1: Alarm occurred |
| L27 | High ambient temperature | BOOL | 0: No alarm; 1: Alarm occurred |
| L28 | High ambient temperature alarm | BOOL | 0: No alarm; 1: Alarm occurred |
| L29 | Low ambient temperature | BOOL | 0: No alarm; 1: Alarm occurred |
| L30 | IGBT module A-phase over-temperature | BOOL | 0: No alarm; 1: Alarm occurred |
| L31 | IGBT module B-phase over-temperature | BOOL | 0: No alarm; 1: Alarm occurred |
| L32 | IGBT module C-phase over-temperature | BOOL | 0: No alarm; 1: Alarm occurred |

### 3.11 Alarm Status 3 (warningGGIO3)

| Param. | Fault Name | Data Type | Description |
|---|---|---|---|
| M01 | IGBT temperature sampling error | BOOL | 0: No alarm; 1: Alarm occurred |
| M02 | IGBT over-temperature alarm | BOOL | 0: No alarm; 1: Alarm occurred |
| M03 | Reactor over-temperature | BOOL | 0: No alarm; 1: Alarm occurred |
| M04 | Reactor over-temperature alarm | BOOL | 0: No alarm; 1: Alarm occurred |
| M05 | Reactor temperature low | BOOL | 0: No alarm; 1: Alarm occurred |
| M06 | Surge protection device fault | BOOL | 0: No alarm; 1: Alarm occurred |
| M07 | Parallel communication fault | BOOL | 0: No alarm; 1: Alarm occurred |
| M08 | Abnormal stop of the master | BOOL | 0: No alarm; 1: Alarm occurred |
| M09 | Carrier synchronization failed | BOOL | 0: No alarm; 1: Alarm occurred |
| M10 | Internal cavity fan fault | BOOL | 0: No alarm; 1: Alarm occurred |
| M11 | Circulating fan fault | BOOL | 0: No alarm; 1: Alarm occurred |
| M12 | Reactor fan fault | BOOL | 0: No alarm; 1: Alarm occurred |
| M13 | IGBT fan fault | BOOL | 0: No alarm; 1: Alarm occurred |
| M14 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| M15 | CAN communication fault between control board and communication board | BOOL | 0: No alarm; 1: Alarm occurred |
| M16 | HMI communication fault | BOOL | 0: No alarm; 1: Alarm occurred |
| M17 | BMS communication fault | BOOL | 0: No alarm; 1: Alarm occurred |

| | | | |
|---|---|---|---|
| M18 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| M19 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| M20 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| M21 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| M22 | EMS communication fault (dual network) | BOOL | 0: No alarm; 1: Alarm occurred |
| M23 | EEPROM fault | BOOL | 0: No alarm; 1: Alarm occurred |
| M24 | EEPROM alarm | BOOL | 0: No alarm; 1: Alarm occurred |
| M25 | Black box error | BOOL | 0: No alarm; 1: Alarm occurred |
| M26 | FPGA data exchange error | BOOL | 0: No alarm; 1: Alarm occurred |
| M27 | RTC error | BOOL | 0: No alarm; 1: Alarm occurred |
| M28 | 220 V control power disconnected | BOOL | 0: No alarm; 1: Alarm occurred |
| M29 | Leakage current Hall self-test failed | BOOL | 0: No alarm; 1: Alarm occurred |
| M30 | Residual abrupt leakage current overlimit | BOOL | 0: No alarm; 1: Alarm occurred |
| M31 | Continuous residual leakage current overlimit | BOOL | 0: No alarm; 1: Alarm occurred |
| M32 | Local emergency stop fault | BOOL | 0: No alarm; 1: Alarm occurred |

### 3.12 Alarm Status 4 (warningGGIO4)

| Param. | Fault Name | Data Type | Description |
|---|---|---|---|
| N01 | Remote emergency stop fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N02 | BMS emergency stop fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N03 | Phase-to-PE fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N04 | Capacitor voltage resonance fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N05 | High humidity | BOOL | 0: No alarm; 1: Alarm occurred |
| N06 | Humidity detection disconnected | BOOL | 0: No alarm; 1: Alarm occurred |
| N07 | Temperature too high | BOOL | 0: No alarm; 1: Alarm occurred |
| N08 | Temperature detection disconnected | BOOL | 0: No alarm; 1: Alarm occurred |
| N09 | Fuse fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N10 | Cabinet door fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N11 | Water pump locked-rotor | BOOL | 0: No alarm; 1: Alarm occurred |
| N12 | Water pump speed regulation signal absent | BOOL | 0: No alarm; 1: Alarm occurred |
| N13 | Water pump idling | BOOL | 0: No alarm; 1: Alarm occurred |
| N14 | Water cooling system hydraulic fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N15 | Fan power sequence incorrect | BOOL | 0: No alarm; 1: Alarm occurred |

| | | | |
|---|---|---|---|
| N16 | Fan power disconnected | BOOL | 0: No alarm; 1: Alarm occurred |
| N17 | Three-phase fan fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N18 | Heater contactor fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N19 | Water pump power disconnected | BOOL | 0: No alarm; 1: Alarm occurred |
| N20 | Software version inconsistent between master and slave | BOOL | 0: No alarm; 1: Alarm occurred |
| N21 | Dynamic environment fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N22 | DC side midpoint sampling disconnected | BOOL | 0: No alarm; 1: Alarm occurred |
| N23 | DC side voltage sampling disconnected | BOOL | 0: No alarm; 1: Alarm occurred |
| N24 | AC and DC voltage mismatch | BOOL | 0: No alarm; 1: Alarm occurred |
| N25 | Grid voltage mismatch | BOOL | 0: No alarm; 1: Alarm occurred |
| N26 | Heat exchanger fan fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N27 | DC circuit breaker fan fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N28 | AC circuit breaker fan fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N29 | Master and slave identification fault | BOOL | 0: No alarm; 1: Alarm occurred |
| N30 | Power inconsistent between master and slave | BOOL | 0: No alarm; 1: Alarm occurred |
| N31 | IGBT temperature low | BOOL | 0: No alarm; 1: Alarm occurred |
| N32 | Zero sequence loop current too large | BOOL | 0: No alarm; 1: Alarm occurred |

### 3.13 Alarm Status 5 (warningGGIO5)

| Param. | Fault Name | Data Type | Description |
|---|---|---|---|
| O01 | FPGA side fault | BOOL | 0: No alarm; 1: Alarm occurred |
| O02 | Fault of parallel devices in cabinet | BOOL | 0: No alarm; 1: Alarm occurred |
| O03 | Fiber communication interrupted | BOOL | 0: No alarm; 1: Alarm occurred |
| O04 | Fiber communication CRC fault | BOOL | 0: No alarm; 1: Alarm occurred |
| O05 | Control mode inconsistent | BOOL | 0: No alarm; 1: Alarm occurred |
| O06 | Capacitor harmonic overlimit | BOOL | 0: No alarm; 1: Alarm occurred |
| O07 | Fan pump and water pump switch error | BOOL | 0: No alarm; 1: Alarm occurred |
| O08 | PCS interaction fault | BOOL | 0: No alarm; 1: Alarm occurred |
| O09 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O10 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O11 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O12 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O13 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O14 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |

| | | | |
|---|---|---|---|
| O15 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O16 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O17 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O18 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O19 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O20 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O21 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O22 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O23 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O24 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O25 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O26 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O27 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O28 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O29 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O30 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O31 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |
| O32 | Reserved | BOOL | 0: No alarm; 1: Alarm occurred |

# 4 GOOSE Data (PIGO)

## 4.1 Remote Measurement Parameters (measGGIO1)

| Param. | Name | Data Type | Description | Lower Limit | Upper Limit | Default | Unit |
|---|---|---|---|---|---|---|---|
| P01 | Rated power | Float32 | - | 0 | 65535 | 1725 | 1 kW |
| P02 | Battery pack SOC | Float32 | - | 0 | 100.0 | 0 | 0.1% |
| P03 | PCS system status–value mode | Int32 | 0: Stopped<br>1: Standby<br>2: On-grid charging operation<br>3: On-grid discharging operation<br>4: On-grid zero power operation<br>5: Off-grid operation<br>6: Fault status | 0 | 65535 | 0 | 1 |
| P04 | Total active output power | Float32 | - | –32768 | 32767 | 0 | 1 kW |
| P05 | Total reactive output power | Float32 | - | –32768 | 32767 | 0 | 1 kVar |
| P06 | Maximum charging power of battery pack | Float32 | - | 0 | 65535 | 0 | 1 kW |
| P07 | Maximum discharging power of battery pack | Float32 | - | 0 | 65535 | 0 | 1 kW |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| P08 | Active power setting | Float32 | Returns actual value; used to determine if setting is successful. | –32768 | 32767 | 0 | 1 kW |
| P09 | Reactive power setting | Float32 | Returns actual value; used to determine if setting is successful. | –32768 | 32767 | 0 | 1 kVar |

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